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Resampling methods

- Bootstrap (i.i.d.)
- Blocking

i.i.d. = independent and identically distributed

$$P(x_1 x_2 \ldots x_n) = \underline{p(x_1)}\, p(x_2) \ldots \underline{p(x_n)}$$

same

mean value

$$\mathbb{E}[x] = \langle x \rangle = \int_{x \in D} P_x(x)\, x\, dx$$

or $$\sum_{i \in D} x_i P_x(x_i)$$

D: $X = \{x_0, x_1 \ldots x_{n-1}\}$

Don't know $P_x(x)$

we evaluate the sample mean $\mu_x = \frac{1}{n} \sum_{i=0}^{n-1} x_i \neq \mathbb{E}[x]$

$$\mathrm{var}[X] = \int_{x \in D} P_X(x)\,(x - \mathbb{E}[X])^2 \, dx$$

$$\left(\sum_i P_X(x_i)\,(x_i - \mathbb{E}[X]) \right)^2$$

$$\langle E^2 \rangle - \langle E \rangle^2$$

$$= \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

in our code we have

standard deviation (STD)

$$\sim \sqrt{\mathrm{var}[X]}$$

sample variance

$$\sigma_x^2 = \frac{1}{n} \sum_{i=0}^{n-1} (x_i - \mu_x)^2$$

$$\neq \mathrm{var}[x]$$

what does it mean that x_i is i.i.d.?

$$p(x_1, x_2) = \tilde{p}(x_1)\, \tilde{p}(x_2)$$

Meet covariance

$$\mathrm{cov}(x_i, x_j) = \int dx_i \int dx_j \, p(x_i, x_j)$$

$$\times (x_i - \bar{x}_i)(x_j - \bar{x}_j)$$

mean value

x_i

$$\bar{x}_i = \frac{1}{n} \sum x_i$$

or $\int x_i \tilde{p}(x_i) \ldots$

if x_i x_j are i.i.d.

$$\bar{x}_i = \bar{x}_j = \bar{x} \wedge p(x_i, x_j) = \tilde{p}(x_i) \times \tilde{p}(x_j)$$

$$\mathrm{cov}(x_i, x_j) =$$

$$\int dx_i \, (x_i - \bar{x}) \, \tilde{p}(x_i) \; \times$$

$$\int dx_j \, (x_j - \bar{x}) \, \tilde{p}(x_j)$$

$$= \langle x_i x_j \rangle - \bar{x}^2 = 0$$

$$\int \tilde{p}(x_i) \, x_i \, dx_i \int \tilde{p}(x_j) \, dx_j \, x_j$$

$$= \bar{x}^2$$

Central limit theorem

- Total mean

$$z = \frac{x_1 + x_2 + \dots + x_m}{m}$$

what is the distribution

$\bar{p}(z)$

we know $p(x_i)$

$$\bar{p}(z) = \int_{x_1 \in D} dx_1\, p(x_1) \int_{x_2 \in D} dx_2\, p(x_2)$$

$$\cdots \int_{x_m \in D} dx_m\, p(x_m)$$

$$\times\, \delta\left(z - \frac{x_1 + x_2 + \cdots x_m}{m}\right)$$

$$\delta\left(z - \frac{x_1 + x_2 + \cdots + x_m}{m}\right)$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} dq\, \exp\left[-iq\left(z - \frac{(x_1 + x_2 + \cdots x_m)}{m}\right)\right]$$

$$\int p(x_i)\, x_i\, dx_i = \mu$$

insert in $\bar{p}(z) =$

$$e^{i\mu q - i\mu q}$$

$$\bar{p}(z) = \frac{1}{2\pi} \int_{-\infty}^{\infty} dq\, e^{iq(z-\mu)}$$

$$\times \left[\underbrace{\int_{x \in D} dx\, p(x) \exp\left[\frac{iq(\mu - x)}{m}\right]} \right]^m$$

$$\int_{x \in D} dx\, p(x) = 1 \quad \wedge \quad \mu = \int dx\, p(x)\, x$$

Taylor-expand [...]

$$\int_{x \in D} p(x)\, dx \left[1 + \frac{i q(\mu - x)}{\hbar} - \frac{q^2 (\mu - x)^2}{2\hbar^2} + \cdots \right]$$

$$= \left[1 + 0 - \frac{q^2}{2\hbar^2} \sigma^2 + \cdots \right]$$

$$\sigma^2 = \int dx\, p(x)\, (x - \mu)^2$$

in the limit $m \to \infty$

$$\bar{p}(z) = \frac{1}{\sqrt{2\pi\sigma^2/m}} \exp\left[-\frac{(z-\mu)^2}{2\sigma^2/m}\right]$$

$p(x)$ has variance σ^2

m experiments lead to

variance $\sigma^2/m \to$

STD $\sim 1/\sqrt{m}$

Bootstrap: first exposure

original sample

$$X = \{x_1, x_2, \ldots x_n\}$$

we assume we can produce new samples X^* and produce new estimates for μ and σ^2

$$\hat{\theta}_n^{*(1)}, \hat{\theta}_n^{*(2)}, \ldots \hat{\theta}_n^{*(B)}$$

$$\theta_n^{*(i)} = \frac{1}{n} \sum_{j=1}^{n} x_j^{*(i)}$$

$$\bar{\theta}_B^{*} = \frac{1}{B} \sum_{i=1}^{B} \theta_n^{*(i)}$$

$$\mathrm{var}\left[\bar{\theta}_B^{*}\right] = \frac{1}{B-1} \sum_{i=1}^{B} \left(\theta_n^{*(i)} - \bar{\theta}_B^{*}\right)^2$$

$$\mathrm{STD} \sim 1 \Big/ \sqrt{B-1}$$

How do we generate $X^{*(i)}$

original

$$X = \{1, 2, 3, 4, 5\}$$

$X^{*(1)}$ by reshuffling randomly with replacement

$$X^{*(1)} = \{2, 3, 3, 5, 1\}$$

Covariance and variance

Total mean of m-experiments

$$\bar{X} = \frac{1}{m} \sum_{i=1}^{m} \bar{x}_i$$

single experiment

$$\bar{x}_i = \frac{1}{n} \sum_{j=1}^{n} x_{ij}$$

Total variance (sample)

$$\sigma_m^2 = \frac{1}{m} \sum_{i=1}^{m} \left(\bar{x}_i - \bar{X}\right)^2$$

$$\bar{x}_i - \bar{X} = \frac{1}{n} \sum_{j=1}^{n} x_{ij} - \bar{X}$$

$$\sigma_m^2 = \frac{1}{m} \sum_{i=1}^{m} \left[\frac{1}{n} \sum_{j=1}^{n} \tilde{x}_{ij}\right]^2$$

$$= \frac{1}{m} \frac{1}{n^2} \sum_{i=1}^{m} \sum_{j=1}^{n} \tilde{x}_{ij}^2 +$$

$$+ \frac{2}{mn^2} \sum_{i=1}^{m} \sum_{j<k}^{n} \tilde{x}_{ij} \tilde{x}_{ik}$$

Total sample variance

$$\sigma^2 = \frac{1}{mn} \sum_{i=1}^{m} \sum_{j=1}^{n} \underbrace{(x_{ij} - \bar{x})^2}_{\tilde{x}_{ij}^2}$$

$$\sigma_m^2 = \frac{\sigma^2}{n} + \underbrace{\frac{2}{mn^2} \sum_{i=1}^{m} \sum_{j<k} \tilde{x}_{ij} \tilde{x}_{ik}}_{\text{2 covariance}}$$

$$d = |j - k|$$

$$\rho_d = \frac{1}{m} \frac{1}{n} \sum_{i=1}^{m} \sum_{k=1}^{n-d} \tilde{x}_{ik} \tilde{x}_{i(k+d)}$$

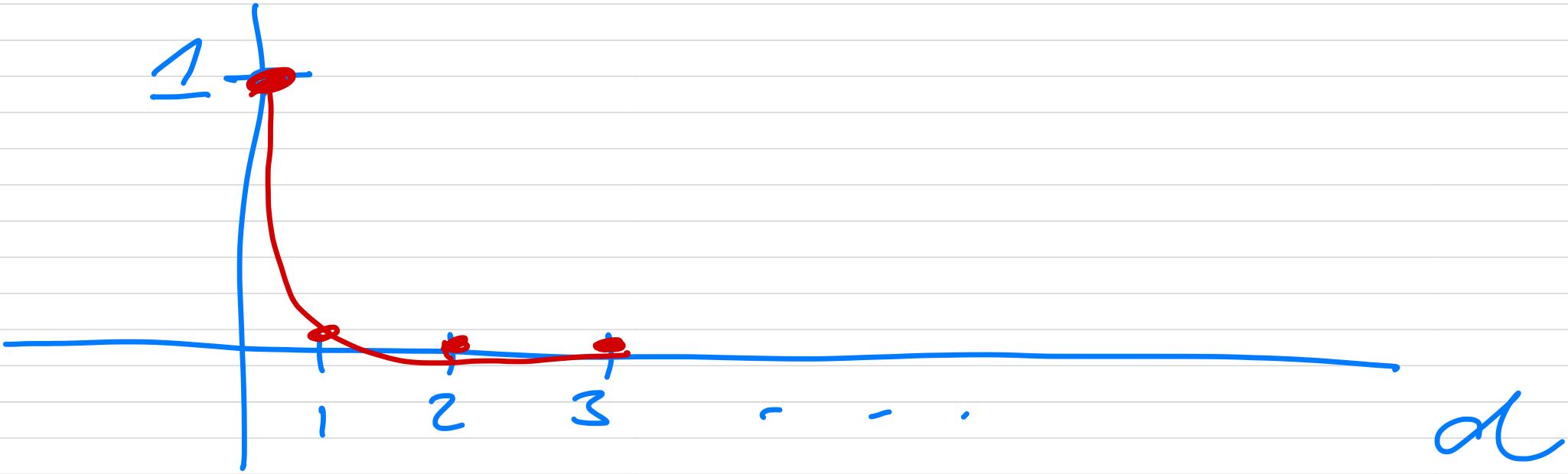
$$d = 0$$

$$\rho_0 = \frac{1}{m} \frac{1}{n} \sum_{i=1}^{m} \sum_{k=1}^{n} \tilde{x}_{ik}$$

$$= \sigma^2$$

$$\sigma_m^2 = \sigma^2/n + \frac{2}{n} \sum_{d=1}^{n-1} \rho_d$$

correlation function

$$K_d = \rho_d / \sigma^2$$

$$d = 0 \Rightarrow K_0 = \sigma^2 / \sigma^2 = 1$$